Introduction to Quantum Physics and Structure of Matter

Lecture 5

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Course Contents

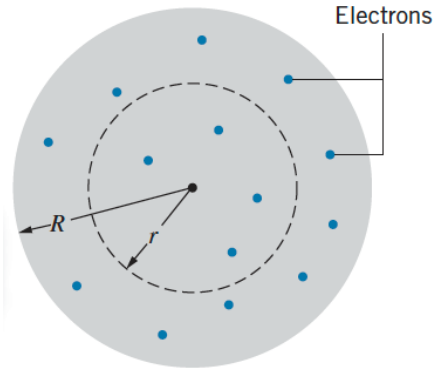
Lecture	Date	Topics	Material
1	03-09-2025	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	10-09-2025	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	17-09-2025	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	24-09-2025	•Basic properties of atoms •Scattering experiments and the Thomson model •The Rutherford nuclear atom •Line spectra •The Bohr model •The Franck-Hertz experiment •Deficiencies of the Bohr model	Book: Ch. 6.1-6.6, 6.8

Course Contents

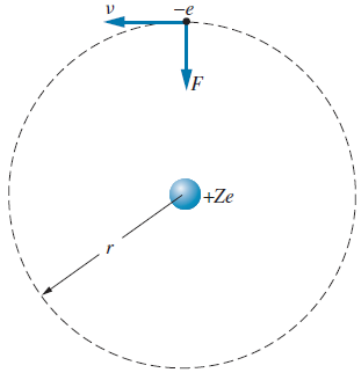
Lecture	Date	Topics	Material
5	**01-10-2025**	**•A one-dimensional atom** **•Angular momentum in the hydrogen atom** **•The hydrogen atom wave functions** **•Radial probability densities** **•Angular probability densities**	**Book: Ch. 7.1-7.7**
6	08-10-2025	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	15-10-2025	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

Recap Previous Lecture

- Atom Models



Thomson



Rutherford

Bohr

- Bohr radii & electron energies (for H-atom)

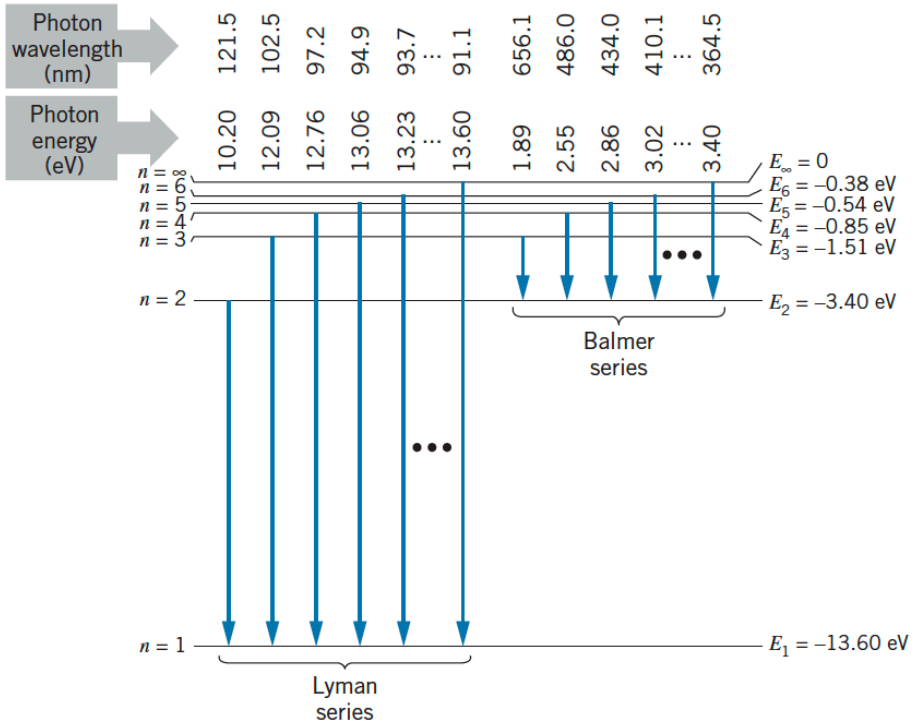
$$r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2 = a_0 n^2$$

$$E_n = -\frac{13.60\ \text{eV}}{n^2}$$

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Recap Previous Lecture

- Energy levels



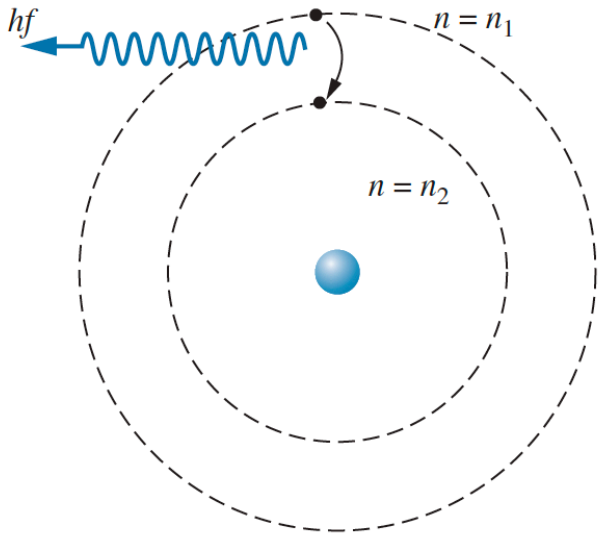
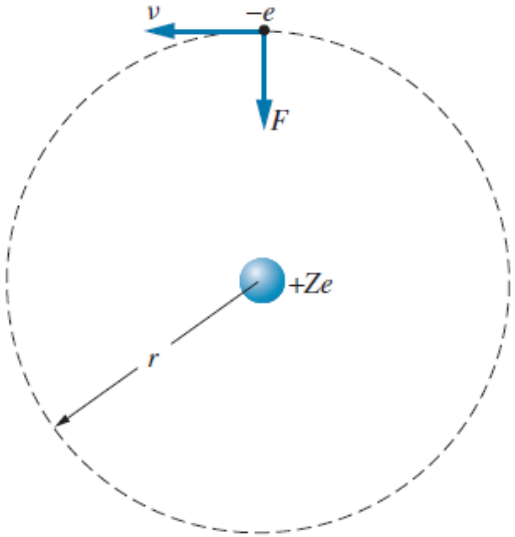


FIGURE 6.19 An electron jumps from the state n_1 to the state n_2 as a photon is emitted.

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Bohr Model vs. Quantum Mechanics



- The Bohr model describes a (classical) circular orbit

- To analyze the H-atom using quantum mechanics, we need to solve the Schrödinger equation with potential:

$$U(r) = -\frac{e^2}{4\pi\epsilon_0 r} \Longrightarrow U(x,y,z) = -\frac{e^2}{4\pi\epsilon_0 r}$$

- This is a 3D-problem, requiring spherical coordinates…

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1D Hydrogen Atom

- Simplify the problem to a 1D atom with: $U(x) = -\frac{e^2}{4\pi\epsilon_0 x}$
- Solving the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \frac{e^2}{4\pi\varepsilon_0 x}\psi(x) = E\psi(x)$$

$x \to \infty \quad \psi(x) \to 0$

$x = 0 \quad \psi(x) = 0$

$$\psi(x) = Axe^{-bx} \qquad b = \frac{1}{a_0} \text{ with } \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

1D Hydrogen Atom

- Substitution of general solution Axe^{-bx} in S.E. :

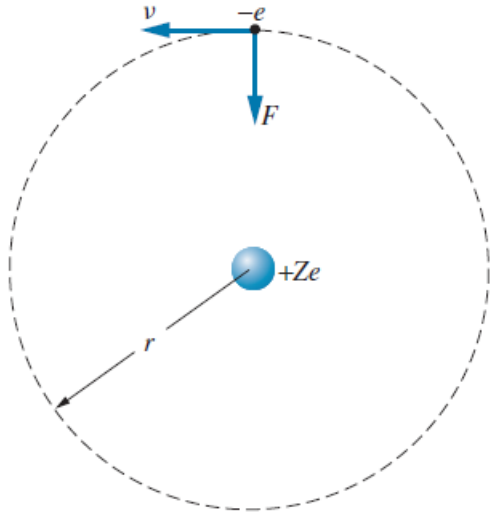
$$\frac{\hbar^2 b}{2m} - \frac{e^2}{4\pi\epsilon_0} = x\left(E - \frac{\hbar^2 b^2}{2m}\right) \rightarrow$$

$$b = \frac{2me^2}{\hbar^2 4\pi\epsilon_0} = \frac{1}{a_0}; \qquad E_n = -\frac{\hbar^2 b^2}{2m} = -\frac{m_e e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -\frac{13.60\ \text{eV}}{n^2}$$

Same energy as Bohr model (a_0: Bohr radius)

Hydrogen Model using Quantum Mechanics

- Assume (Bohr model) a (classical) circular orbit



- To analyze the H-atom using quantum mechanics, we need to solve the Schrödinger equation with (electrical) potential:

$$U(r) = -\frac{e^2}{4\pi\epsilon_0 r} \Longrightarrow U(x,y,z) = -\frac{e^2}{4\pi\epsilon_0 r}$$

- This is a 3D-problem, requiring spherical coordinates…

1D Hydrogen Atom

- Simplify the problem to a 1D atom with: $U(x) = -\frac{e^2}{4\pi\epsilon_0 x}$
- Solving the 1D Schrödinger equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \frac{e^2}{4\pi\varepsilon_0 x}\psi(x) = E\psi(x)$$

Boundary conditions :

$$\left.\begin{array}{c} \psi(\infty) \to 0 \\ \lim_{x\downarrow 0}(\psi(x)/x) \neq \infty:\ \psi(0) = 0 \end{array}\right\} \text{General solution } \psi(x) = Axe^{-bx}$$

1D Hydrogen Atom

Substitution of Axe^{-bx} in S.E. :

$$\frac{\hbar^2 b}{2m} - \frac{e^2}{4\pi\epsilon_0} = x\left(E - \frac{\hbar^2 b^2}{2m}\right) \rightarrow$$

$$b = \frac{2me^2}{\hbar^2 4\pi\epsilon_0} = \frac{1}{a_0}; \qquad E_n = -\frac{\hbar^2 b^2}{2m} = -\frac{m_e e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -\frac{13.60\ \text{eV}}{n^2}$$

Same energy as Bohr model (a_0: Bohr radius)

1D Hydrogen Atom

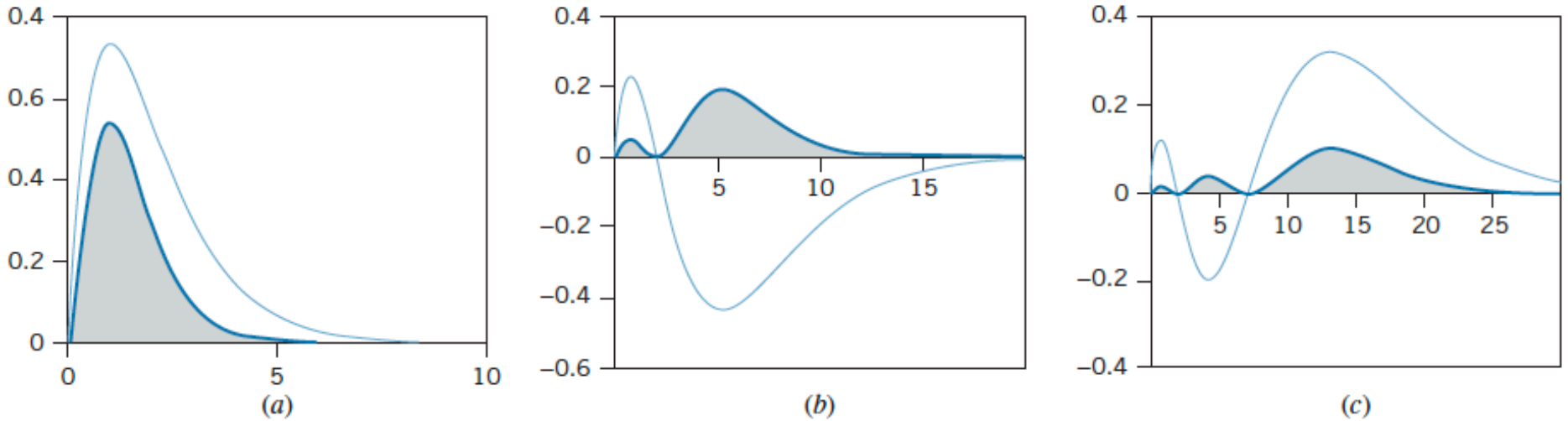


FIGURE 7.1 Wave functions and probability densities (shaded areas) for an electron bound in a one-dimensional Coulomb potential energy. The horizontal axis represents the distance between the proton and electron in units of a_0. (a) Ground state. (b) First excited state. (c) Second excited state.

Unlike the Bohr model, there is a non-zero probability for the electron between $0 < x < \infty$, instead of just at $r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2 = a_0 n^2$

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1D Hydrogen Atom – Exercise

We have found the solution for a particle trapped in a 1D Coulomb potential:

$$\psi(x) = Axe^{-bx} \qquad b = \frac{1}{a_0} \text{ with } \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

Calculate A using the normalization constraint, i.e. $\int_0^\infty \left|\psi(x)^2\right| dx = 1$

Hint: use the standard integral $\int_0^\infty x^n e^{-cx} dx = \frac{n!}{c^{n+1}}$

1D Hydrogen Atom – Exercise

$\psi(x) = Axe^{-bx}$ $\quad b = \frac{1}{a_0}$ with $\quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$ $\quad \int_0^\infty x^n e^{-cx} dx = \frac{n!}{c^{n+1}}$

$$\int_0^\infty |\psi(x)|^2 dx = A^2 \int_0^\infty x^2 e^{-\frac{2x}{a_0}} dx = 1$$

With $n = 2$ and $c = 2/a_0$:

$$A^2 \frac{2!}{\left(\frac{2}{a_0}\right)^3} = A^2 \frac{2a_0^3}{2^3} = 1 \implies A = \sqrt{\frac{2^3}{2a_0^3}} = 2a_0^{-\frac{3}{2}}$$

The hydrogen atom: 3D Schrödinger Equation

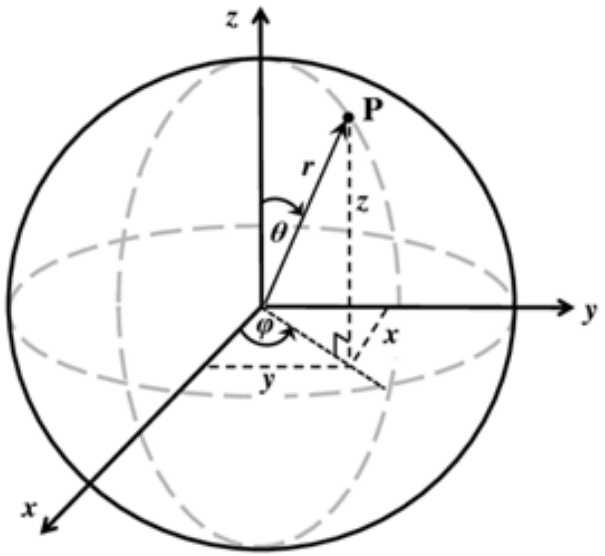
$$-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z) + U(x,y,z)\psi(x,y,z) = E\psi(x,y,z)$$

Coulomb Potential: $U(x,y,z) = -\frac{e^2}{4\pi\epsilon_0}\frac{1}{\sqrt{x^2+y^2+z^2}} = -\frac{e^2}{4\pi\epsilon_0}\frac{1}{r}$

As potential is function of radius, use spherical coordinates:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(r,\phi,\theta) + U(r,\phi,\theta)\psi(r,\phi,\theta) = E\psi(r,\phi,\theta)$$

3D Schrödinger Equation



$x = r \sin\theta \cos\varphi$

$y = r \sin\theta \sin\varphi$

$z = r \cos\theta$

$r = \sqrt{x^2 + y^2 + z^2}$

$\theta = \cos^{-1}(z/r)$

$\varphi = \tan^{-1}(y/x)$

$$d\vec{\mathbf{r}} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\,\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$$

$$\vec{\mathbf{F}} = F_r\hat{\mathbf{r}} + F_\theta\hat{\boldsymbol{\theta}} + F_\phi\hat{\boldsymbol{\phi}}$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$$

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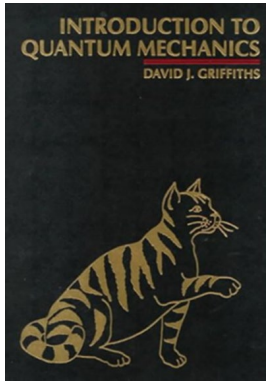
3D Schrödinger Equation

The Schrödinger equation in spherical coordinates:

$$-\frac{\hbar^2}{2m}\left[\frac{\partial^2\psi}{\partial r^2}+\frac{2}{r}\frac{\partial\psi}{\partial r}+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right]+U\psi=E\psi$$

U only dependent on r: write $\psi(r,\theta,\varphi)=R(r)\Theta(\theta)\Phi(\varphi)$

with: $R(r)$ the radial function
$\Theta(\theta)\Phi(\varphi)$ the angular function



With $\psi(r,\theta,\psi) = R(r)\Theta(\theta)\Phi(\varphi)$ (shorthand: $\psi(r,\theta,\psi) = R\Theta\Phi$)

$$-\frac{\hbar^2}{2m}\left[\frac{Y}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{R}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right)+\frac{R}{r^2\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}\right]+U\,R\Theta\Phi = E\,R\Theta\Phi$$

Divide by $R\Theta\Phi$ and multiply by $-2mr^2/\hbar^2$:

$$\left\{\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-\frac{2mr^2}{\hbar^2}[U(r)-E]\right\} = -\frac{1}{\Theta\Phi}\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta\Phi}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2\Theta\Phi}{\partial\varphi^2}\right\}$$

Radial part — Angular part

Expressions left & right must be constant, call this constant $l(l+1)$

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Angular function solution: spherical harmonics, function of parameters m, l

For constant $l(l+1)$ we find various solutions $\Phi(\theta, m, l)$ and $\Theta(\theta, m, l)$ where $m \leq l$

l	m_l	$\Theta(\theta)$	$\Phi(\phi)$
0	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
0	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
1	0	$\sqrt{\frac{3}{2}}\cos\theta$	$\frac{1}{\sqrt{2\pi}}$
1	±1	$\mp\frac{\sqrt{3}}{2}\sin\theta$	$\frac{1}{\sqrt{2\pi}}e^{\pm i\phi}$
0	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
1	0	$\sqrt{\frac{3}{2}}\cos\theta$	$\frac{1}{\sqrt{2\pi}}$
1	±1	$\mp\frac{\sqrt{3}}{2}\sin\theta$	$\frac{1}{\sqrt{2\pi}}e^{\pm i\phi}$
2	0	$\sqrt{\frac{5}{8}}(3\cos^2\theta - 1)$	$\frac{1}{\sqrt{2\pi}}$
2	±1	$\mp\sqrt{\frac{15}{4}}\sin\theta\cos\theta$	$\frac{1}{\sqrt{2\pi}}e^{\pm i\phi}$
2	±2	$\frac{\sqrt{15}}{4}\sin^2\theta$	$\frac{1}{\sqrt{2\pi}}e^{\pm 2i\phi}$

Radial function

The radial component $R(r)$:

$$-\frac{\hbar^2}{2m}\left(\frac{d^2R}{dr^2}+\frac{2}{r}\frac{dR}{dr}\right)+\left(-\frac{e^2}{4\pi\epsilon_0 r}+\frac{l(l+1)\hbar^2}{2mr^2}\right)R(r)=ER(r)$$

With $u(r) \coloneqq r\,R(r)$ this becomes

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2}+\left(U+\frac{\hbar^2}{2m}\,\frac{l(l+1)}{r^2}\right)u=Eu$$

This is the 1-dimensional schrodinger equation again, with $U_{eff} = U+\frac{\hbar^2}{2m}\,\frac{l(l+1)}{r^2}$, i.e. U plus a *centrifugal term* that 'throws the electron outwards' .

In hydrogen (Coulomb potential) we can calculate E:

$$E = -\frac{me^4}{32\,\pi^2\epsilon_0^2\hbar^2}\,\frac{1}{n^2}$$

n : *principal quantum number*

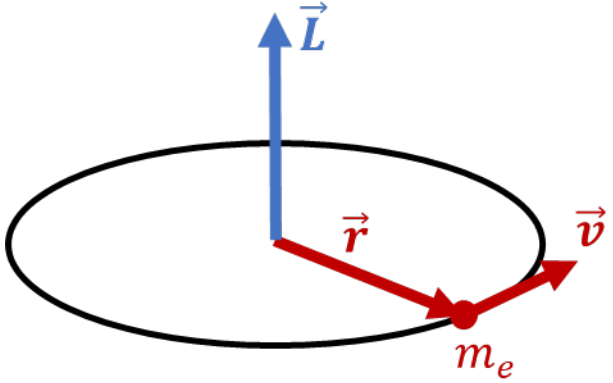
We can show that for given n, l is restricted to values <n:

$l = 0,1,2,\ldots,n-1$ *angular momentum quantum number*

$m = 0,\pm1,\pm2,\ldots\pm l$ *magnetic quantum number*

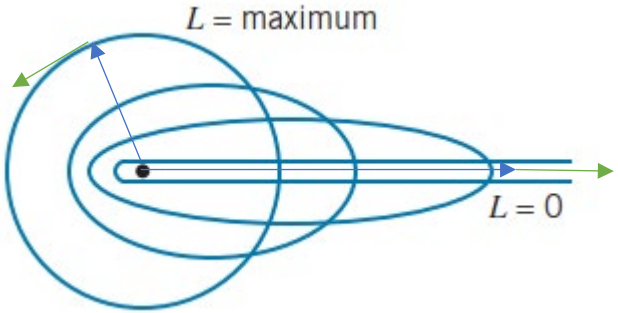
Solving the Schrodinger equation automatically introduces these quantum numbers

Angular Momentum – Classical



- Classical angular momentum: $\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}} = \vec{\boldsymbol{r}} \times (m_e \vec{\boldsymbol{v}})$

- To describe the full plane, we need all three components: $\vec{\boldsymbol{L}}(L_x, L_y, L_z)$, or give two angular coordinates and magnitude ($|\vec{\boldsymbol{L}}|$)



Same energy, different L

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Angular Momentum – Quantum Mechanics

- The angular momentum properties of a three-dimensional wave function are described by two quantum numbers:

1. Angular momentum quantum number l:

$$\left|\vec{\boldsymbol{L}}\right| = \sqrt{l(l+1)}\hbar \qquad \text{with} \quad l = 0,1,2,\ldots$$

2. Magnetic quantum number m_l:

$$L_z = m_l\hbar \qquad \text{with} \quad m_l = 0, \pm 1, \pm 2, \ldots, \pm l$$

3. ??? → Related to uncertainty principle (we will discuss this later)

Note that for each value of l there are $2l + 1$ possible values of m_l

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Angular Momentum – Exercise

What are the possible z-components of the vector $\vec{\boldsymbol{L}}$ that represents the orbital angular momentum of a state with $l = 2$? And what is the length of this vector?

$$\left|\vec{\boldsymbol{L}}\right| = \sqrt{l(l+1)}\hbar \qquad \text{with} \quad l = 0,1,2,\ldots$$

$$L_z = m_l\hbar \qquad \text{with} \quad m_l = 0, \pm 1, \pm 2, \ldots, \pm l$$

Angular Momentum – Exercise

$l = 2$:

$$\left|\vec{\boldsymbol{L}}\right| = \sqrt{l(l+1)}\hbar = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar$$

$$L_z = m_l\hbar \qquad \text{with} \quad m_l = 0, \pm 1, \pm 2, \ldots, \pm l$$

So, we can have $m_l = -2, -1, 0, 1, 2$ (5 possible components)

$$L_z = -2\hbar, -\hbar, 0, \hbar, 2\hbar$$

Angular Momentum – Quantum Mechanics

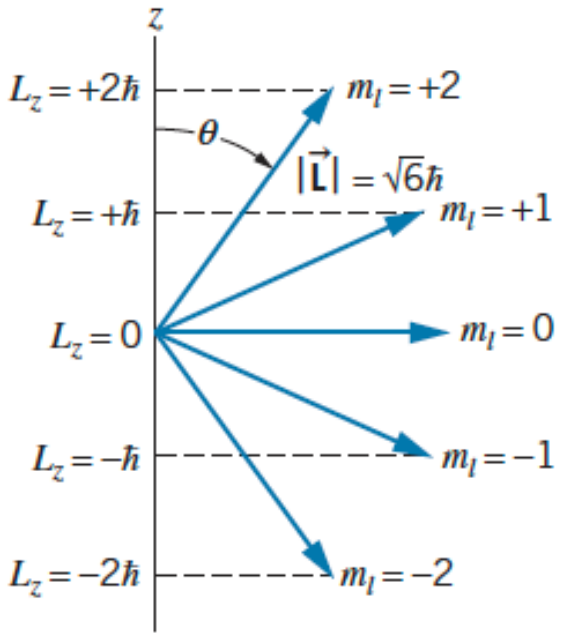


FIGURE 7.3 The orientations in space and z components of a vector with $l = 2$. There are five different possible orientations.

$$\cos\theta = \frac{L_z}{\left|\vec{\boldsymbol{L}}\right|} = \frac{m_l}{\sqrt{l(l+1)}}$$

Number of possible states:

$$2l + 1$$

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Angular Momentum & Uncertainty Principle

$$\Delta L_z \Delta\phi \geq \hbar$$

We can only reduce the uncertainty in $\Delta\phi$ (L_x, L_y) at the cost of increasing the uncertainty of ΔL_z

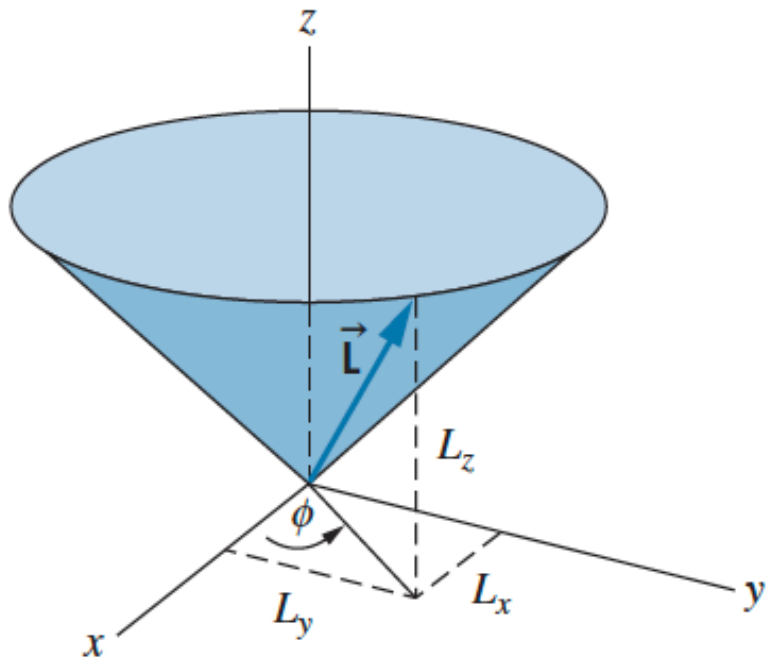


FIGURE 7.4 The vector $\vec{\mathbf{L}}$ precesses rapidly about the z axis, so that L_z stays constant, but L_x and L_y are indeterminate.

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TABLE 7.1 Some Hydrogen Atom Wave Functions

n	l	m_l	$R(r)$	$\Theta(\theta)$	$\Phi(\phi)$
1	0	0	$\frac{2}{a_0^{3/2}} e^{-r/a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
2	0	0	$\frac{1}{(2a_0)^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
2	1	0	$\frac{1}{\sqrt{3}(2a_0)^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\sqrt{\frac{3}{2}} \cos\theta$	$\frac{1}{\sqrt{2\pi}}$
2	1	± 1	$\frac{1}{\sqrt{3}(2a_0)^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$	$\mp \frac{\sqrt{3}}{2} \sin\theta$	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$
3	0	0	$\frac{2}{(3a_0)^{3/2}} \left(1 - \frac{2r}{3a_0} + \frac{2r^2}{27a_0^2}\right) e^{-r/3a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
3	1	0	$\frac{8}{9\sqrt{2}(3a_0)^{3/2}} \left(\frac{r}{a_0} - \frac{r^2}{6a_0^2}\right) e^{-r/3a_0}$	$\sqrt{\frac{3}{2}} \cos\theta$	$\frac{1}{\sqrt{2\pi}}$
3	1	± 1	$\frac{8}{9\sqrt{2}(3a_0)^{3/2}} \left(\frac{r}{a_0} - \frac{r^2}{6a_0^2}\right) e^{-r/3a_0}$	$\mp \frac{\sqrt{3}}{2} \sin\theta$	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$
3	2	0	$\frac{4}{27\sqrt{10}(3a_0)^{3/2}} \frac{r^2}{a_0^2} e^{-r/3a_0}$	$\sqrt{\frac{5}{8}} (3\cos^2\theta - 1)$	$\frac{1}{\sqrt{2\pi}}$
3	2	± 1	$\frac{4}{27\sqrt{10}(3a_0)^{3/2}} \frac{r^2}{a_0^2} e^{-r/3a_0}$	$\mp \sqrt{\frac{15}{4}} \sin\theta \cos\theta$	$\frac{1}{\sqrt{2\pi}} e^{\pm i\phi}$
3	2	± 2	$\frac{4}{27\sqrt{10}(3a_0)^{3/2}} \frac{r^2}{a_0^2} e^{-r/3a_0}$	$\frac{\sqrt{15}}{4} \sin^2\theta$	$\frac{1}{\sqrt{2\pi}} e^{\pm 2i\phi}$

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Wave Functions Hydrogen – Exercise

How many different (n, l, m_l) states are there for hydrogen with $n = 3$?

$$\psi_{n,l,m_l}(r, \theta, \varphi) = R_{n,l}(r)\Theta_{l,m_l}(\theta)\Phi_{m_l}(\varphi)$$

With $n = 1,2,3, \ldots$ $l = 0,1,2, \ldots, n-1$ $m_l = 0, \pm 1, \pm 2, \ldots, \pm l$

Wave Functions Hydrogen – Exercise

How many different (n, l, m_l) states are there for hydrogen with $n = 3$?

$$\psi_{n,l,m_l}(r, \theta, \varphi) = R_{n,l}(r)\Theta_{l,m_l}(\theta)\Phi_{m_l}(\varphi)$$

With $n = 1,2,3, \ldots$ $l = 0,1,2, \ldots, n-1$ $m_l = 0, \pm 1, \pm 2, \ldots, \pm l$

$n = 3, l = 0$	$n = 3, l = 1$	$n = 3, l = 2$
3,0,0	3,1,-1	3,2,-2
	3,1,0	3,2,-1
	3,1,1	3,2,0
		3,2,1
		3,2,2

9 different states for same energy

Wave Functions Hydrogen

- The energy levels are given as: $E_n = -\frac{1}{(4\pi\varepsilon_0)^2}\frac{me^4}{2n^2\hbar^2} = -\frac{13.6}{n^2}\ \text{eV}$
  - Energy level only depends on n, not on l or m_l
  - Energy levels are the same as in the Bohr model
  - There are different sets of quantum numbers with the same energy

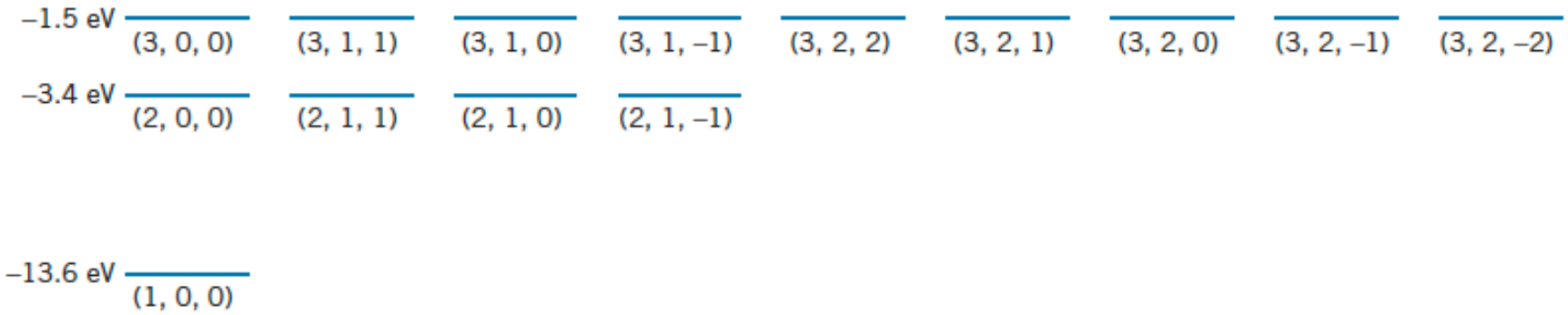


FIGURE 7.6 The lower energy levels of hydrogen, labeled with the quantum numbers (n, l, m_l). The first excited state is four-fold degenerate and the second excited state is nine-fold degenerate.

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Wave Functions Hydrogen – Probabilities

$$\psi_{n,l,m_l}(r,\theta,\varphi) = R_{n,l}(r)\Theta_{l,m_l}(\theta)\Phi_{m_l}(\varphi)$$

$\left|\psi_{n,l,m_l}(r,\theta,\varphi)\right|^2$ → *volume probability density*

$$\left|\psi_{n,l,m_l}(r,\theta,\varphi)\right|^2 dV = \left|R_{n,l}(r)\right|^2 \left|\Theta_{l,m_l}(\theta)\right|^2 \left|\Phi_{m_l}(\varphi)\right|^2 \underbrace{r^2 \sin\theta \, dr \, d\theta \, d\phi}_{dV}$$

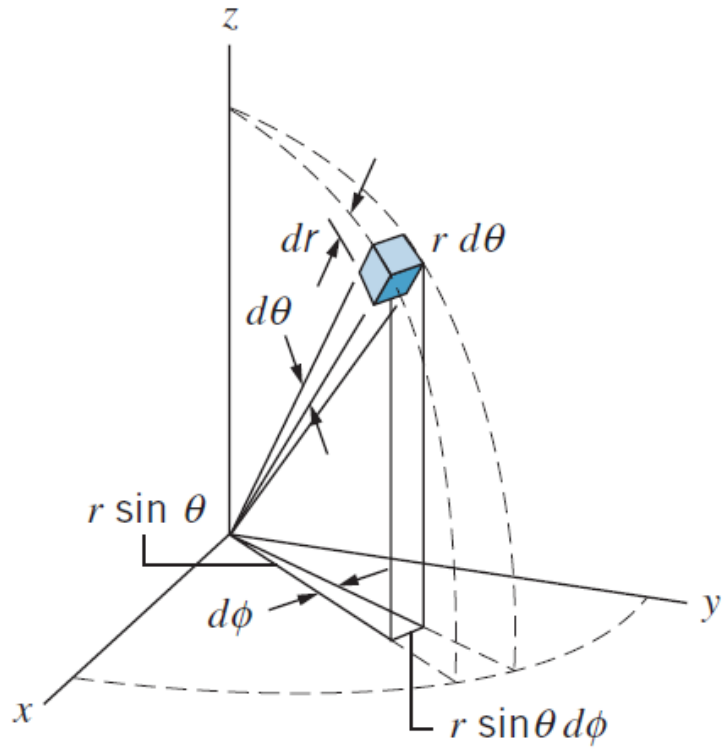


FIGURE 7.8 The volume element in spherical polar coordinates.

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Wave Functions Hydrogen – Probabilities

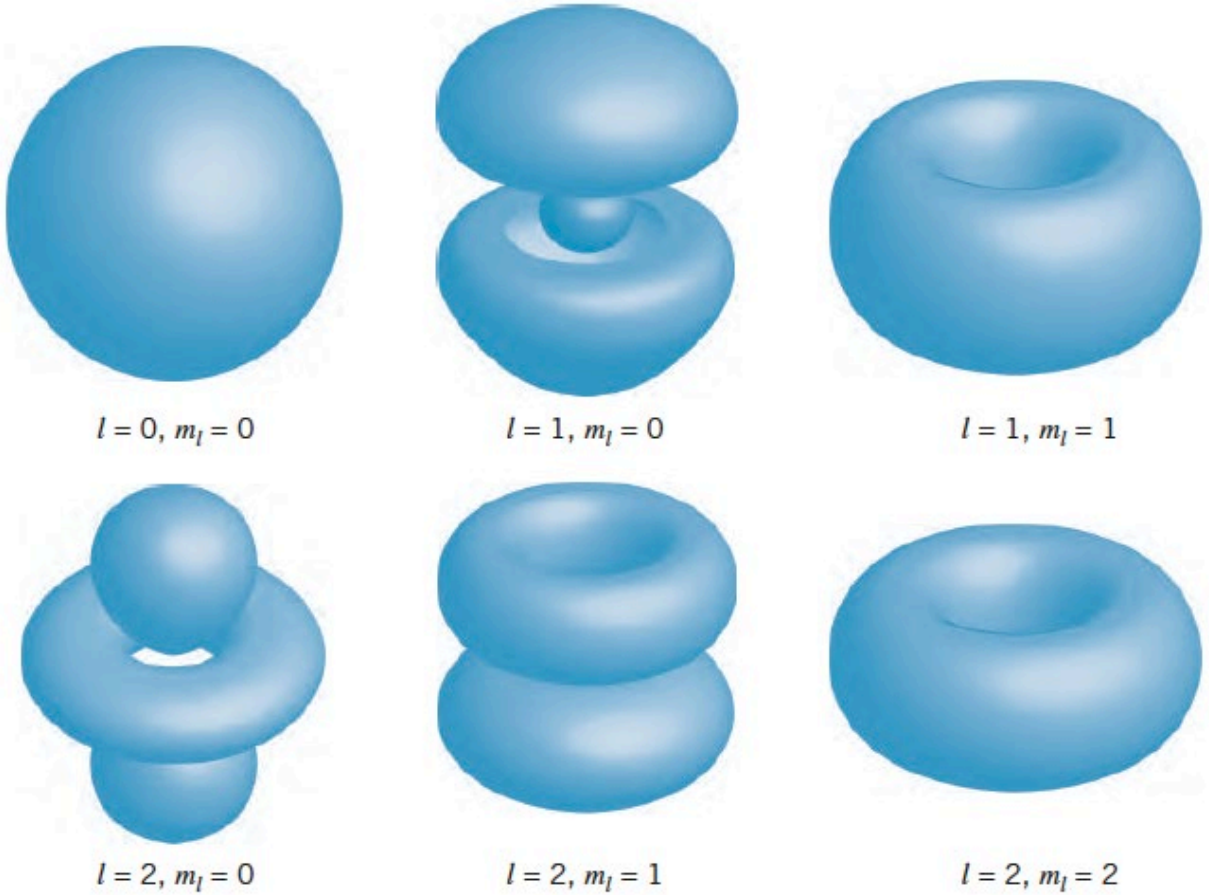


FIGURE 7.9 Representations of $|\psi|^2$ for different sets of quantum numbers. The z axis is the vertical direction. The diagrams represent surfaces on which the probability has the same value.

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Wave Functions Hydrogen – Probabilities

- Instead of finding the complete probability to locate an electron, we might want to focus only on the probability of a specific distance from the nucleus:

$$P(r)dr = \left|R_{n,l}(r)\right|^2 r^2 dr \underbrace{\int_0^{\pi} \left|\Theta_{l,m_l}(\theta)\right|^2 \sin\theta \, d\theta}_{1} \underbrace{\int_0^{2\pi} \left|\Phi_{m_l}(\varphi)\right|^2 d\phi}_{1}$$

Radial probability density

$$P(r) = r^2 \left|R_{n,l}(r)\right|^2$$

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Wave Functions Hydrogen – Probabilities

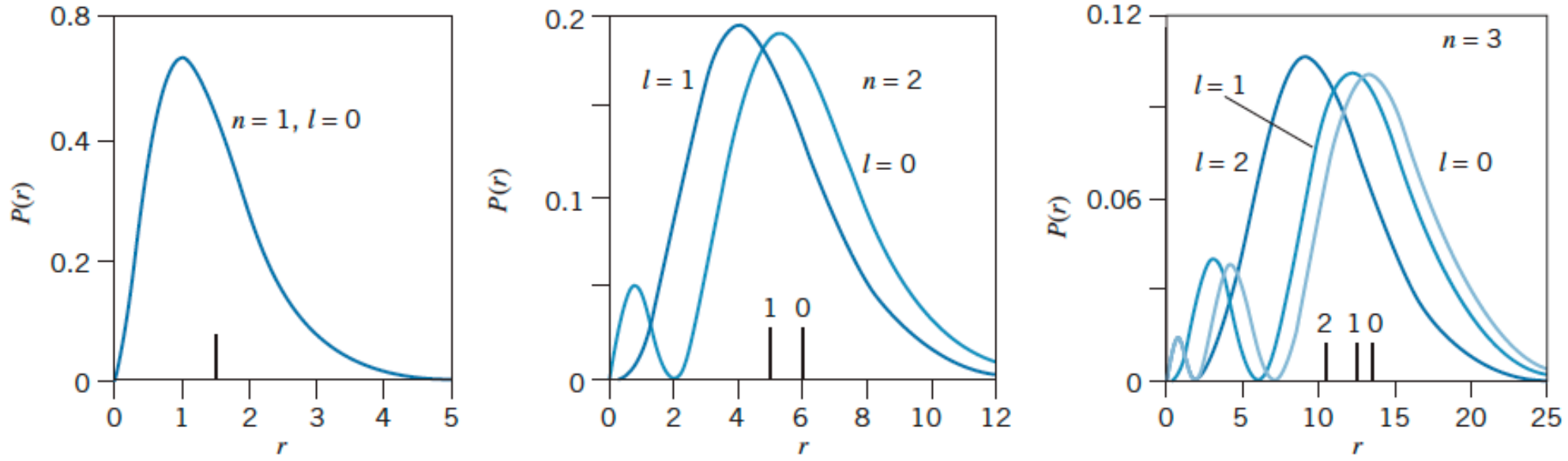


FIGURE 7.10 The radial probability density $P(r)$ for the $n = 1, n = 2$, and $n = 3$ states of hydrogen. The radius coordinate is measured in units of a_0. The markers on the horizontal axis show the values of the average radius r_{av} labeled with the value of l.

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Wave Functions Hydrogen – Probabilities

Angular probability density

$$P(\theta,\phi) = \left|\Theta_{l,m_l}(\theta)\Phi_{m_l}(\varphi)\right|^2$$

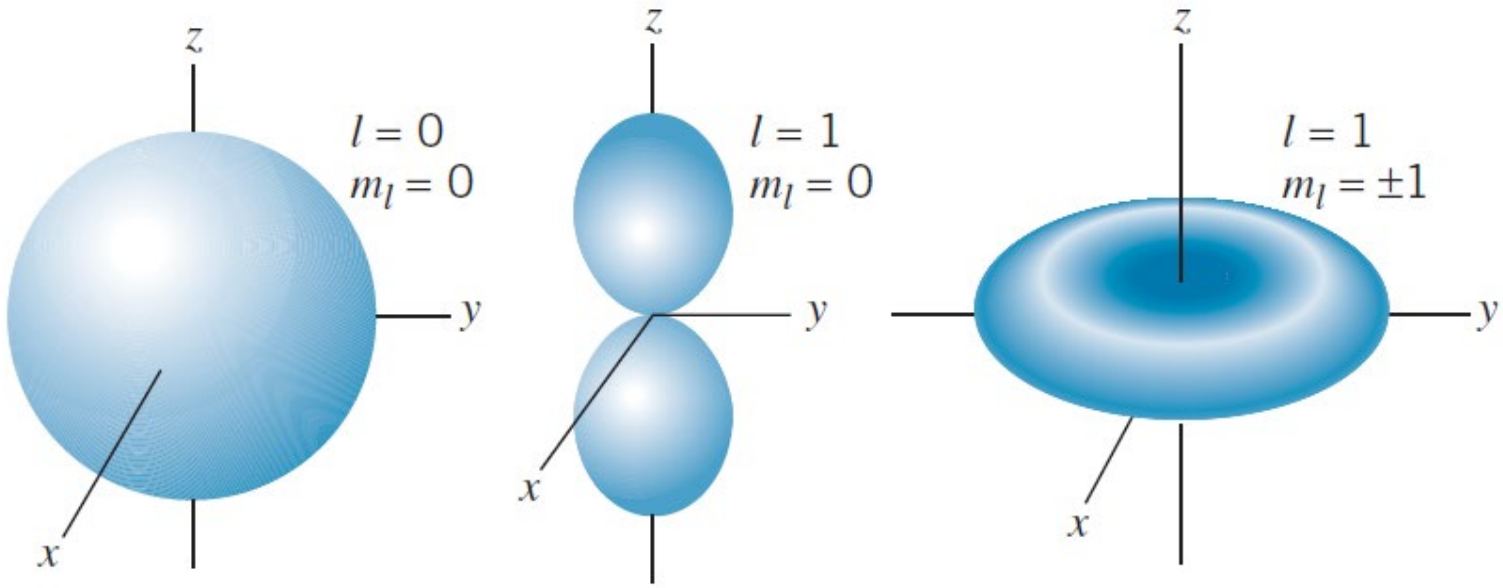


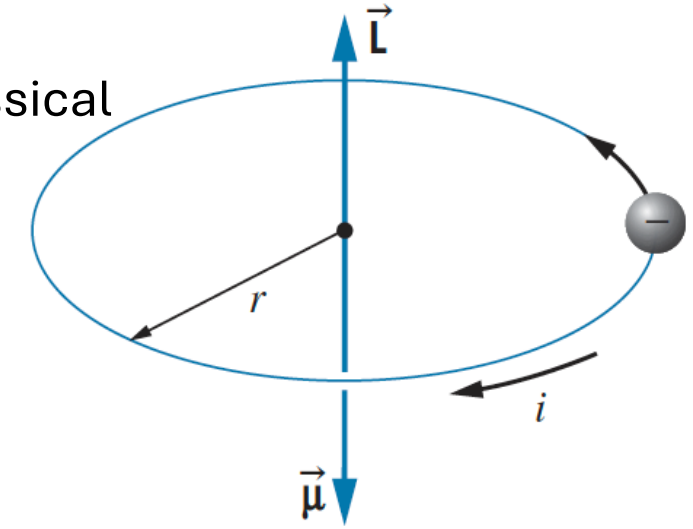
FIGURE 7.11 The angular dependence of the $l = 0$ and $l = 1$ probability densities.

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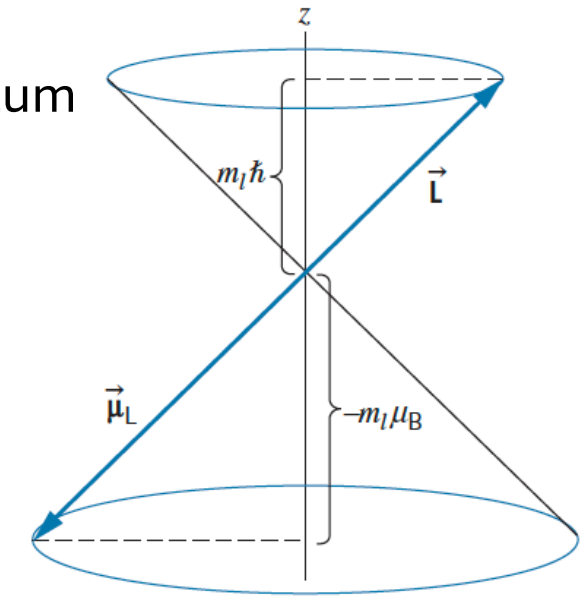
Magnetic Dipole Moment

- Classical: dipole moment defined as $\vec{\mu} = I\,\vec{A}$; aligns with external magnetic field due to Lorentz force on moving charge ('current')
- To observe the *spatial quantization* in experiments, external magnetic fields are used
- Goal: observe special orientation of $\vec{L}$ (components/m_l number)

Classical



Quantum



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Magnetic Dipole Moment

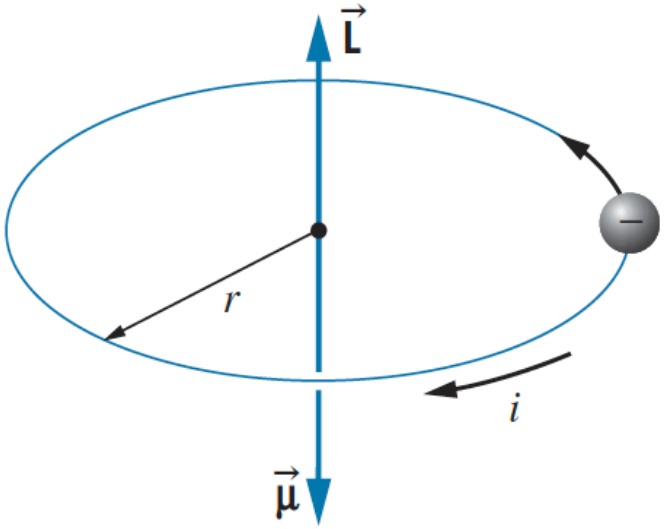
$$i = \frac{dq}{dt} = \frac{q}{T}$$

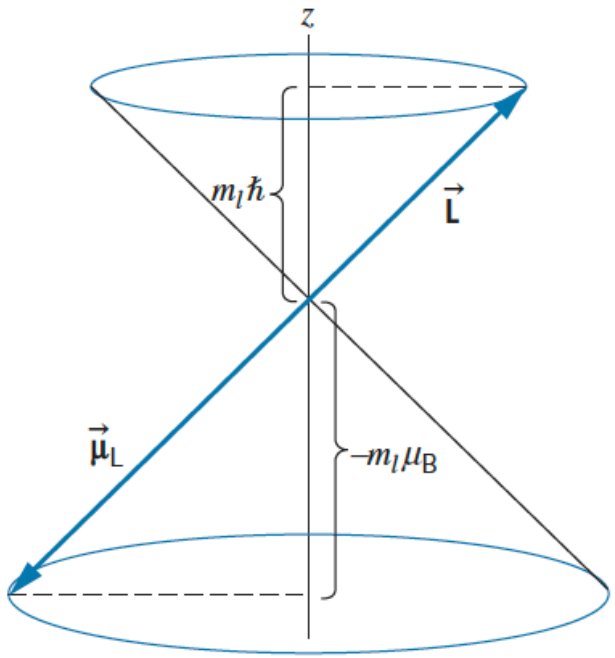
$$T = \frac{2\pi r}{v} = \frac{2\pi r m}{p}$$

$$\mu = iA = \frac{q}{\frac{2\pi r m}{p}}\,\pi r^2 = \frac{q}{2m} rp = \frac{q}{2m}\left|\vec{L}\right|$$

$$\overrightarrow{\mu_L} = -\frac{e}{2m}\vec{L}$$

So: having an angular moment ($\vec{L}$) means there is a magnetic dipole moment ($\overrightarrow{\mu_L}$)

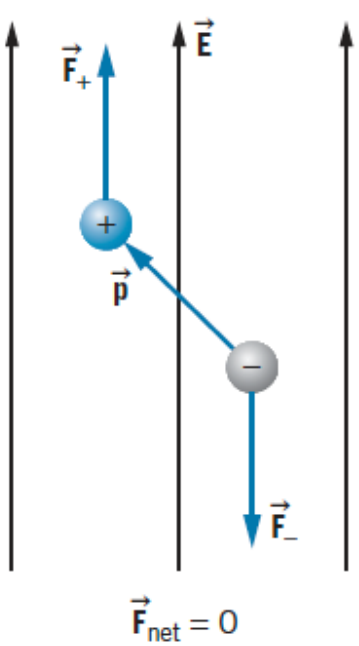


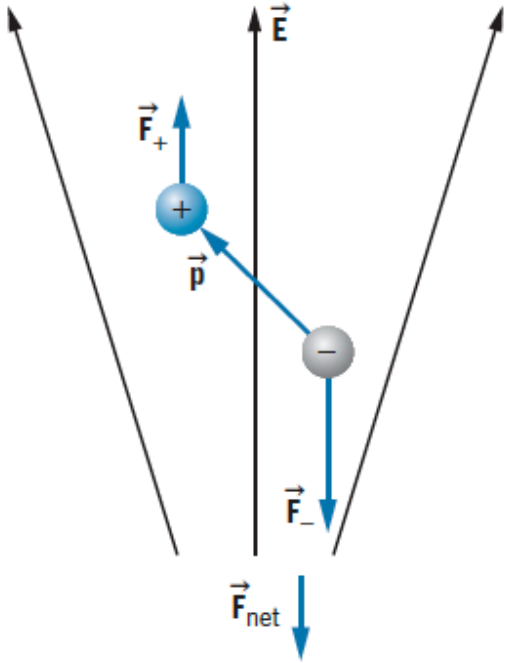


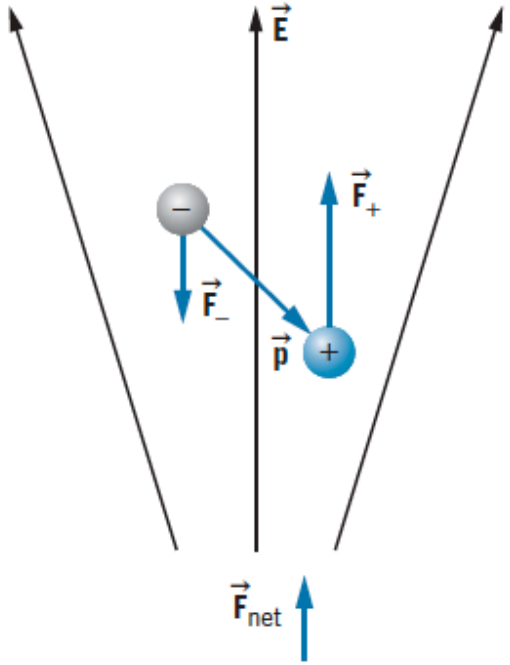
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Magnetic Dipole Moment

- A magnetic dipole can be moved in an external field
- Analogy with electric dipoles:







Magnetic Dipole Moment

- A magnetic dipole can be moved in an external field

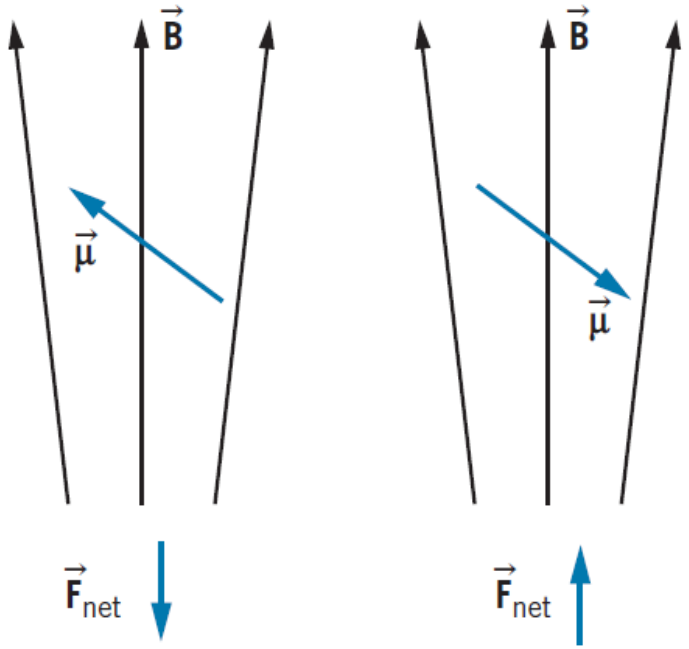
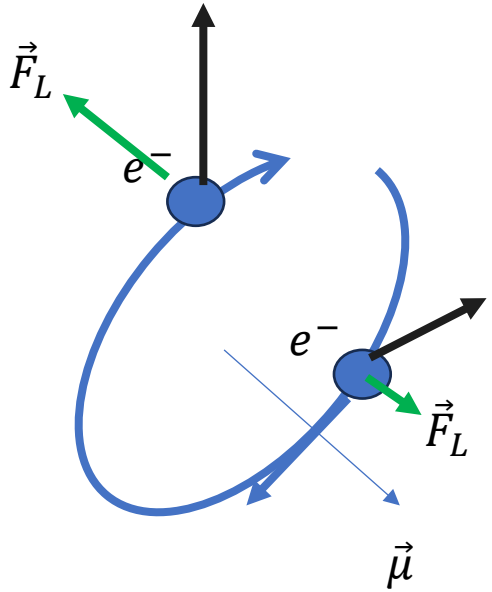


FIGURE 7.15 Two magnetic dipoles in a nonuniform magnetic field. Oppositely directed dipoles experience net forces in opposite directions.



In homogeneous B field: Lorentz forces cancel out in orbit (although there is a net torque forcing the orbit to a path ⊥ to B);
In *inhomogeneous* B field: Lorentz forces do not cancel and the electron/atom moves (dependent on original orbit position, i.e. m_l:
We expect different trajectories for every value of m_l

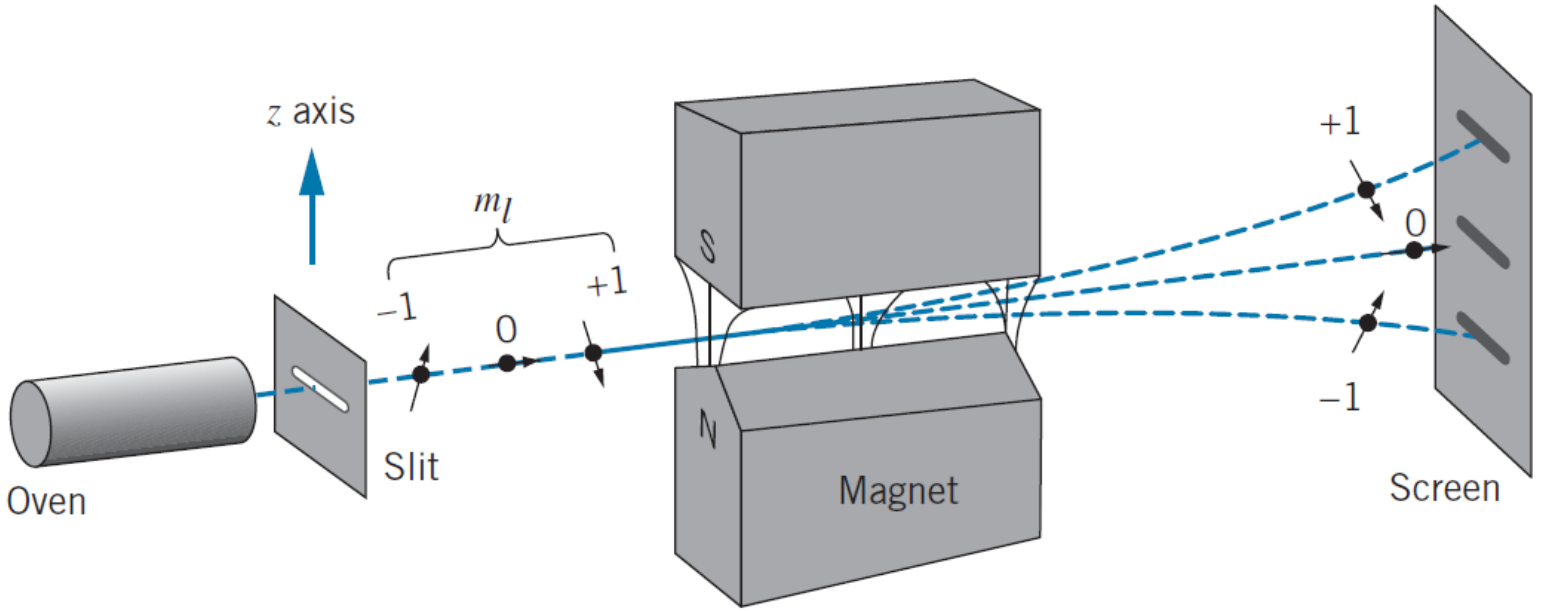
Charge experiences $\vec{F}_L = q\vec{v} \times \vec{B}$

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Stern-Gerlach Experiment

Otto Stern
1888-1969





- Beam of silver atoms in $n = 2, l = 1$ state
- Equal number of atoms in $m_l = -1,0,+1$ states
- Beam goes through non-uniform magnetic field
- For $l = 2$: $m_l = -2,-1,0,+1,+2$ states → 5 images
- Since $m_l = 0, \pm 1, \ldots., \pm\, l$, we should always get odd number of images $(2l + 1)$

Stern-Gerlach Experiment

- Results:
  - a) $B = 0$: No "smeared out" images but spatial quantization ✓
  - b) $B \neq 0$: An even number of images is shown,
    even in the ground state $l = 1$: $l = \frac{1}{2}$ is not possible... ?

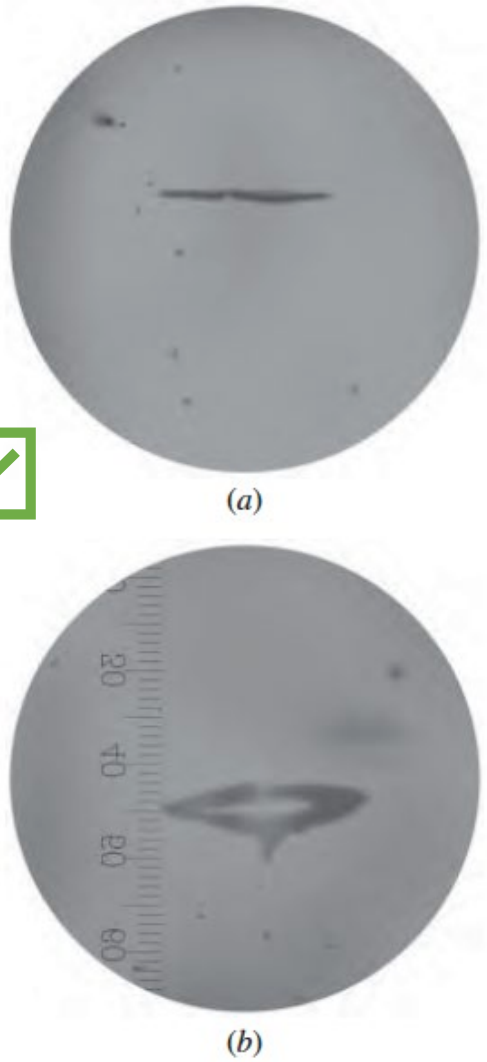


FIGURE 7.17 The results of the Stern-Gerlach experiment. (*a*) The image of the slit with the field turned off. (*b*) With the field on, two images of the slit appear. The small divisions in the scale at the left represent 0.05 mm. [*Source:* W. Gerlach and O. Stern, *Zeitschrift für Physik* **9**, 349 (1922)]

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Intrinsic Angular Momentum

- An electron has 2 kinds of angular momentum:
  1. *Orbital* angular momentum:
     motion of the electron around the nucleus

  2. *Intrinsic* angular momentum or *spin*, which behaves much like the orbital angular momentum

- Spin quantum number: $s = \frac{1}{2}$

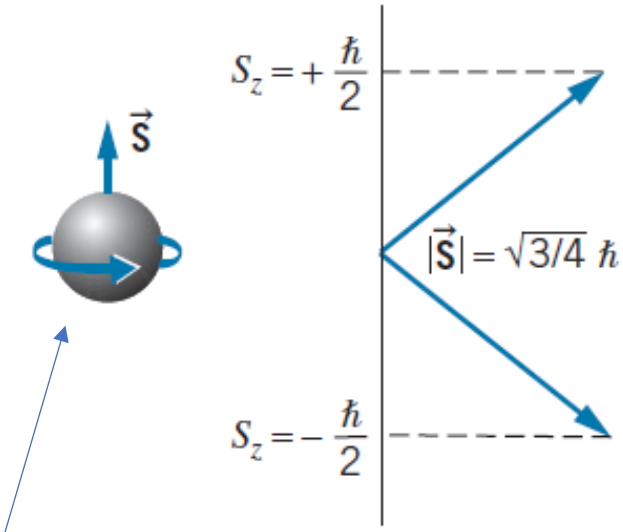


FIGURE 7.18 The spin angular momentum of an electron and the spatial orientation of the spin angular momentum vector.

! Actually misrepresentation as electron has no discernible size

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Spin Quantum Number

- Spin angular momentum: vector $\vec{\boldsymbol{S}}$ with length $|\vec{\boldsymbol{S}}| = \sqrt{s(s+1)}\hbar$ where *s* is the spin quantum number

- For an electron $s = \frac{1}{2}$, so $|S| = \sqrt{3/4}\hbar$

- The *z*-component of $\vec{S}$ can have two values:

$$S_z = m_s\hbar \text{ with } m_s = \pm\frac{1}{2}$$

where m_s is the *magnetic spin quantum number*

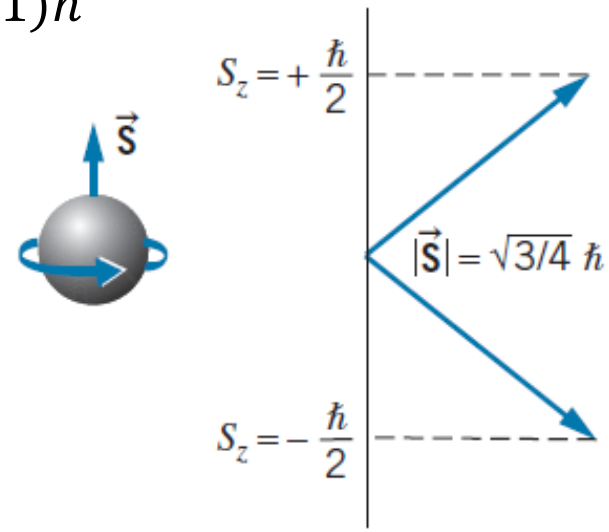


FIGURE 7.18 The spin angular momentum of an electron and the spatial orientation of the spin angular momentum vector.

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Orbital vs. Spin Angular Momentum

TABLE 7.2 Orbital and Spin Angular Momentum of Electrons in Atoms

	Orbital	Spin
Quantum number	$l = 0, 1, 2, \ldots$	$s = 1/2$
Length of vector	$\lvert\vec{\mathbf{L}}\rvert = \sqrt{l(l+1)}\hbar$	$\lvert\vec{\mathbf{S}}\rvert = \sqrt{s(s+1)}\hbar = \sqrt{3/4}\hbar$
z component	$L_z = m_l\hbar$	$S_z = m_s\hbar$
Magnetic quantum number	$m_l = 0, \pm1, \pm2, \ldots, \pm l$	$m_s = \pm 1/2$
Magnetic moment	$\vec{\mu}_L = -(e/2m)\vec{\mathbf{L}}$	$\vec{\mu}_S = -(e/m)\vec{\mathbf{S}}$

Quantum Numbers

The hydrogen atom is described by four quantum numbers:
n, l, m_l, m_s

n	principal quantum number,
l	angular momentum quantum number,
m_l	magnetic quantum number
m_s	magnetic spin quantum number

Spectroscopic Notation

- When an atom is not in a magnetic field, the values for m_l and m_s are of no significance for its energy level

- In this case, we use spectroscopic notation for the energy levels:

Value of l	0	1	2	3	4	5	6
Designation	s	p	d	f	g	h	i

- Transitions rule: $\Delta l = \pm 1$, according to Schrödinger

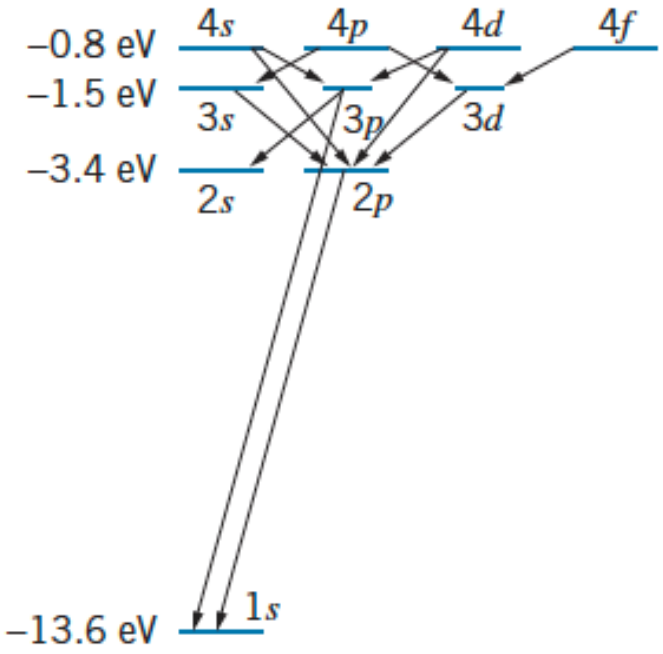


FIGURE 7.19 A partial energy level diagram of hydrogen, showing the spectroscopic notation of the levels and some of the transitions that satisfy the $\Delta l = \pm 1$ selection rule.

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